Asymmetric Catalysis

How to cite: *Angew. Chem. Int. Ed.* **2022,** *61,* e202211635
International Edition: doi.org/10.1002/anie.202211635
German Edition: doi.org/10.1002/ange.202211635

# Rhodium-Catalyzed Dynamic Kinetic [4+2] Cycloaddition of Allene-1,3-Dienes

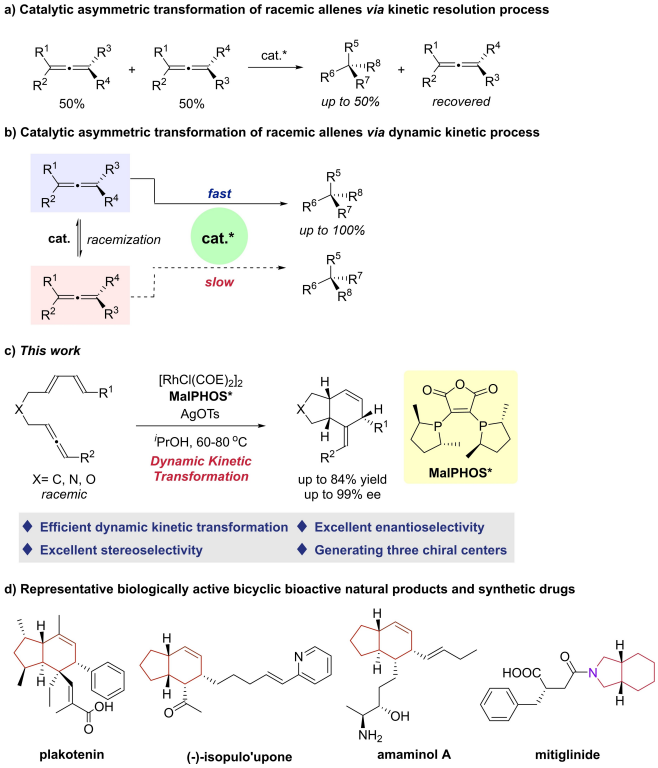
*Yulin Han[+], Anni Qin[+], Qian Zhang, Xue Zhang,* Hui Qian,* and Shengming Ma**

*In memory of Prof. Jirō Tsuji*

**Abstract:** Rh-catalyzed dynamic kinetic intramolecular [4+2] cycloaddition reaction of 1,3-disubstituted allene-1,3-dienes afforded *cis*-fused [4.3.0]bicyclic products with an excellent chemo-, diastereo-, and enantioselectivity. Many synthetically useful functional groups are tolerated. The synthetic utility has been demonstrated. Based on the careful experimental studies, a mechanism involving the rapid racemization of the allene moiety in the starting materials has been proposed.

## Introduction

Allenes are versatile building blocks[1] and have been becoming an integral part in organic synthesis due to their unique axial chirality[2] and chemical properties.[3] Although some examples of classic kinetic resolution (KR) with the theoretic maximum yield of 50% have been reported (Scheme 1a),[4] catalytic dynamic kinetic transformations involving the racemization of allenes have not been well established (Scheme 1b).[5–8] Only a few examples of catalytic allene racemization have been reported.[9] Thus, establishing overall processes combining both the rapid racemization and excellent enantioselective conversions are still of high interest. Herein, we wish to report the first example of rhodium-catalyzed dynamic kinetic transformation of racemic allene-1,3-dienes, i.e., the asymmetric [4+2] cycloaddition of allene-1,3-dienes[10] with Malphos[11] as the chiral ligand, affording *cis*-fused [4.3.0]bicyclic products, which are important skeletons in bioactive compounds[12] (Scheme 1d). It should be noted that three chiral centers have been generated at the same time with an excellent enantio- and diastereoselectivity (Scheme 1c).




***Scheme 1.*** Asymmetric synthesis involving racemic allenes and representative [4.3.0]bicyclic molecules.

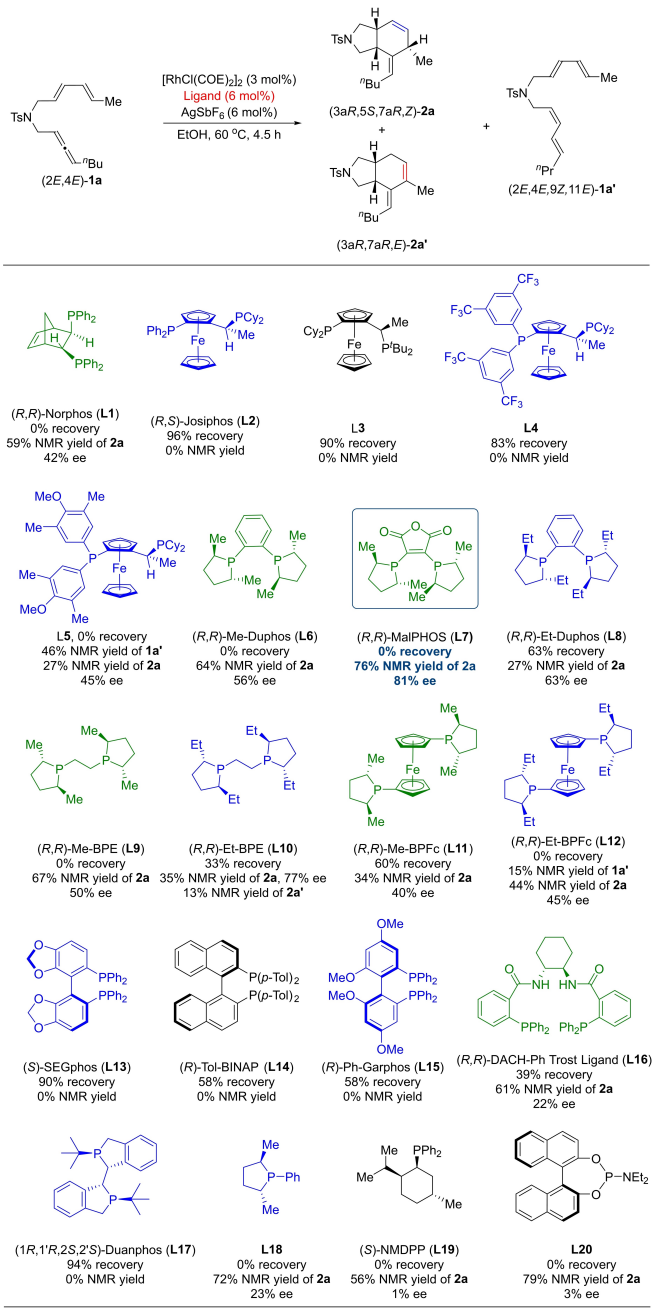
## Results and Discussion

We commenced our study using racemic 1,3-disubstituted allene-1,3-diene (2*E*,4*E*)-**1a** as the model substrate with $[RhCl(COE)_2]_2$ (3 mol%, COE = *cis*-cyclooctene) and $AgSbF_6$ (6 mol%) to identify the best ligand for the reaction in ethanol at 60 °C (Scheme 2). (*R*,*R*)-Norphos **L1** was initially employed as the ligand, product (3a*R*,5*S*,7a*R*,*Z*)-**2a** was successfully obtained in 59% yield but with only 42% ee; ferrocene-based phosphine ligands **L2**–**L4** did not give any products; Josiphos ligand with electron-donating group **L5** could afford product (3a*R*,5*S*,7a*R*,*Z*)-**2a** in 27% yield

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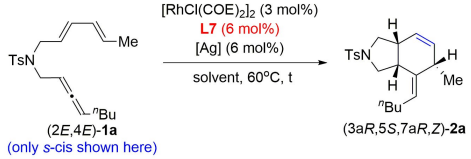

***Scheme 2.*** Ligand screening. Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a**, 3 mol% of $[RhCl(COE)_2]_2$, 6 mol% of $AgSbF_6$, and 6 mol% of ligand in EtOH (1 mL) at 60 °C. NMR yields were presented determined by $^1H$ NMR analysis and ee was determined by HPLC analysis. BPE = 1,2-Bisphospholanoethane; BPFc = 1,1′-Bisphospholanoferrocene; NMDPP = Neomenthyldiphenylphosphine.

with 45% ee, 46% of byproduct tetraene (2*E*,4*E*,9*Z*,11*E*)-**1a′** was also obtained;[4e] (*R*,*R*)-Me-Duphos **L6** slightly improved the yield to 64% and ee to 56%; surprisingly, when (*R*,*R*)-Malphos **L7**, which has a larger bite angle than (*R*,*R*)-Me-Duphos, was used, the enantioselectivity was 81% with a yield of 76%. Other chiral bisphosphines having diverse two-carbon bridges **L8**–**L12** could not give better results. We also investigated other biarylphosphine ligands. (*S*)-SEGphos **L13**, (*R*)-Tol-BINAP (**L14**, BINAP = 1.1′-Binaphthyl-2.2′-diphemyl phosphine), (*R*)-Ph-Garphos **L15** and Duanphos **L17** were inefficient for this reaction. (*R*,*R*)-DACH-Ph Trost Ligand (**L16**, DACH = diaminocyclohexane) was efficient in producing the product (3a*R*,5*S*,7a*R*,*Z*)-**2a**, however, the enantioselectivity is very low. Monophosphine ligands **L18**–**L20** afforded (3a*R*,5*S*,7a*R*,*Z*)-**2a** in an essentially racemic form. Based on these results of ligand screening, Malphos is the most efficient ligand. It may be due to its perfect bite angle (86.1°) as compared with those of **L6** (83.6°) and **L9** (83.2°), **L1** (123°), **L11** (96°), and **L16** (125°).

With (*R*,*R*)-Malphos as the ligand, other reaction parameters were then studied (Table 1): Firstly, various silver salts were screened, and AgOTs was found to be optimal with 51% of yield and an ee of 90% (Entries 1–3). When the silver salt loading was increased to 7.0 mol%, 69% yield of product could be obtained with 86% ee (Entry 4). Lower yields and enantioselectivity of (3*a*R,5-*S*,7*a*R,*Z*)-**2a** was obtained when the reaction was performed in MeOH and HFIP (hexafluoroisopropanol) under otherwise identical conditions (Entries 5 and 6). $^i$PrOH turned out to be the optimal, allowing the formation of the product in 65% yield with 95% ee (Entry 7). When the reaction was conducted at 80 °C, the yield of **2a** could be improved to 81% (78% isolated yield) with an ee of 95% within just 0.5 h (Entry 8). Thus, the optimal conditions have been defined as 3 mol% of $[RhCl(COE)_2]_2$, 6 mol% of (*R*,*R*)-Malphos, and 7 mol% of AgOTs in $^i$PrOH at 80 °C.

With the optimized reaction conditions in hand, the scope has been investigated (Scheme 3). Allene-dienes (2*E*,4*E*)-**1a**–**1k** bearing 1°-alkyl ($R^2$) group ranging from Me

***Table 1:*** Optimization of the reaction conditions.[a]



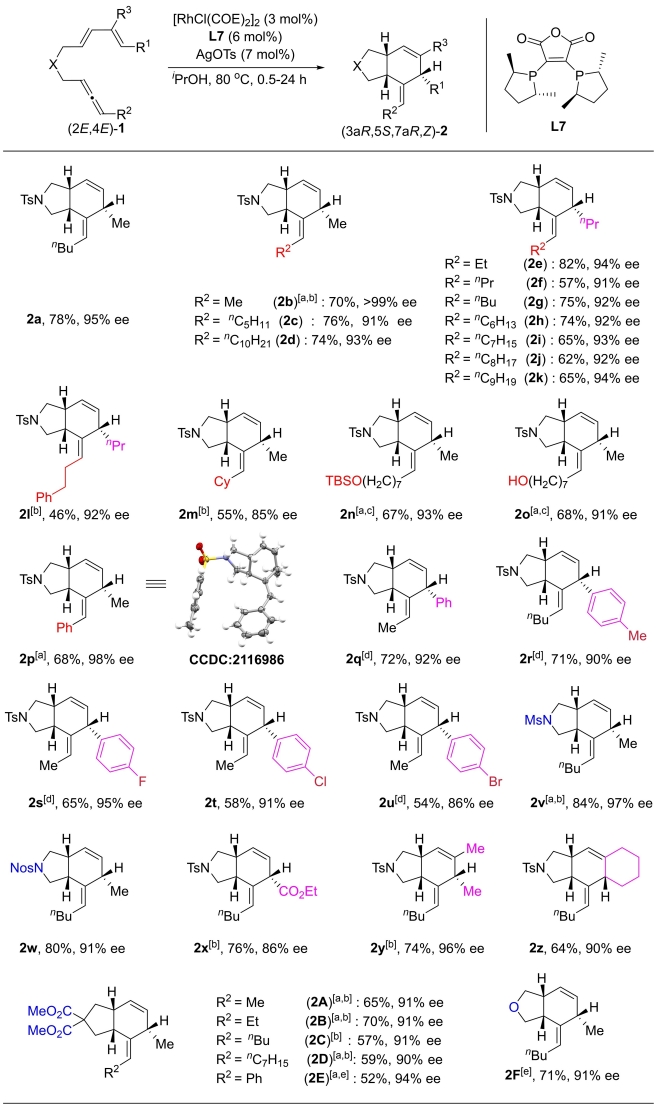

| Entry | [Ag] | solvent | t (h) | (3a*R*,5*S*,7a*R*,*Z*)-**2a** Yield[b] (%) | ee[c] (%) | Recovery of **1a** (%) |
|---|---|---|---|---|---|---|
| 1 | $AgBF_4$ | EtOH | 6 | 36 | 83 | 24 |
| 2 | AgOMs | EtOH | 4.5 | 56 | 76 | 0 |
| 3 | AgOTs | EtOH | 9 | 51 | 90 | 0 |
| 4[d] | AgOTs | EtOH | 6 | 69 | 86 | 0 |
| 5[d] | AgOTs | MeOH | 4.5 | 35 | 84 | 40 |
| 6[d] | AgOTs | HFIP | 6 | 27 | 40 | 0 |
| 7[d] | AgOTs | $^i$PrOH | 12 | 65 | 95 | 0 |
| 8[d,e] | AgOTs | $^i$PrOH | 0.5 | 81(78) | 95 | 0 |

[a] Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a**, 3 mol% of $[RhCl(COE)_2]_2$, 6 mol% of [Ag], and 6 mol% of **L7** in solvent (1 mL) at 60 °C. [b] Determined by $^1H$ NMR analysis using mesitylene as the internal standard. [c] Determined by HPLC analysis. [d] 7 mol% of [Ag]. [e] The reaction was conducted at 80 °C.

**NOTE: The "Ligand*" in the equation of Table 1 should also be revised as "**L7**". Please check the additional DOC file named "Revised Table 1"**


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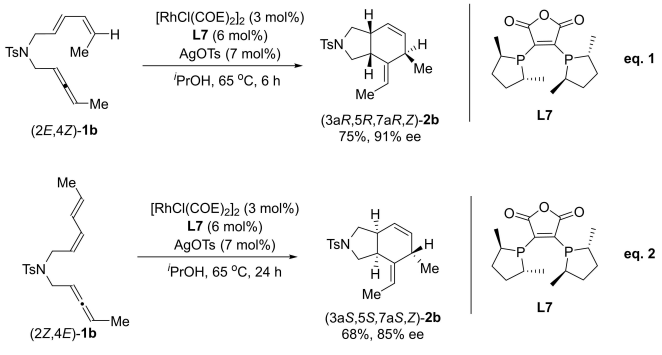

***Scheme 3.*** Scope study. Reaction conditions: 0.2 mmol of (2*E*,4*E*)-**1**, 3 mol% of $[RhCl(COE)_2]_2$, 7 mol% of AgOTs and 6 mol% of **L7** in $^i$PrOH (2 mL) at 80 °C. Isolated yields were presented and ee was determined by HPLC analysis. [a] 0.5 mmol of (2*E*,4*E*)-**1** was used. [b] The reaction was conducted at 60 °C. [c] The reaction was conducted at 65 °C. [d] $^i$PrOH (1.5 mL) and toluene (0.5 mL). [e] The reaction was conducted at 70 °C.

to $^n$C$_{10}$H$_{21}$ all afforded products **2a**–**2k** with excellent ee in moderate to good yields; the reaction afforded the corresponding product **2l** in 46% yield and 92% ee with R$^2$ being phenethyl; cyclohexyl-substituted **1m** also worked with an excellent ee for the formation of **2m**. It is worth mentioning that this alkyl chain may be substituted with silyl-protected alcohol and free hydroxyl group, affording the corresponding products **2n** and **2o**; R$^2$ may also be a phenyl group (**2p**) with its absolute configurations of the chiral centers being established via its *X*-ray diffraction study.[13] The reaction afforded corresponding [4+2] cycloaddition products with a decent efficiency when R$^1$ group was alkyl or phenyl. Phenyl groups substituted with electron-donating (Me) and electron-withdrawing versatile F, Cl, or Br can also be tolerated (**2q**–**2u**). The reaction is amenable to electron-withdrawing ester group (**2x**). When the diene is substituted with more than one substituent, they also reacted smoothly affording corresponding cyclic products **2y** and **2z** with a decent efficiency. The allenes and 1,3-dienes may also be tethered with nitrogen (**2v**–**2w**), malonate (**2A**–**2E**), and oxygen (**2F**).

Furthermore, the reactions of the C=C bond stereoisomers (2*E*,4*Z*)-**1b** and (2*Z*,4*E*)-**1b** under the optimal conditions afforded the diastereoisomers (3a*R*,5*R*,7a*R*,*Z*)-**2b** (75% yield, 91% ee) (Scheme 4, eq 1) and (3a*S*,5*S*,7a*S*,*Z*)-**2b** (68% yield, 85% ee), respectively (Scheme 4, eq 2), indicating that the absolute configurations of the product were determined by the C=C bond configurations of the diene moiety and the chiral ligand.

To demonstrate the practicality of this protocol, 1.0785 g of (2*E*,4*E*)-**1a** was submitted to the standard reaction conditions to afford 81% yield of (3a*R*,5*S*,7a*R*,*Z*)-**2a** with 97% ee with 2.0 mol% of the catalyst (Scheme 5a). The bicylic product (3a*R*,5*S*,7a*R*,*Z*)-**2a** can be transformed into different versatile products (Scheme 5b): Ts (*p*-toluenesulfonyl) group could be removed in the presence of sodium naphthalene to afford **3a** without racemization; Epoxidation with 1.1 equiv *m*CPBA (*m*-chloroperoxybenzoic acid) afforded the single epoxide **4a** with an excellent diastereoselectivity. The absolute configurations of the chiral centers in **4a** were established via its *X*-ray diffraction study;[13] Dihydroxylation with $OsO_4$ afforded diol product **5a** with an excellent dr; Tricyclic product **6a** was easily prepared by Diels–Alder reaction with **2a** and 3,6-dichloro-1,2,4,5-tetrazine; Furthermore, aryl bromide substituted products **2u** could react with (3-nitrophenyl)boronic acid to give the Suzuki coupling product **2u″** in 68% yield with >99% ee. Furthermore, (2*E*,4*E*)-**1G** was prepared and submitted to the standard reaction conditions. The reaction afforded 78% yield and 95% ee of (3a*R*,5*S*,7a*R*,*Z*)-**2G**, which is the structural analogue of (−)-isopulo′upone (Scheme 5c).[14]

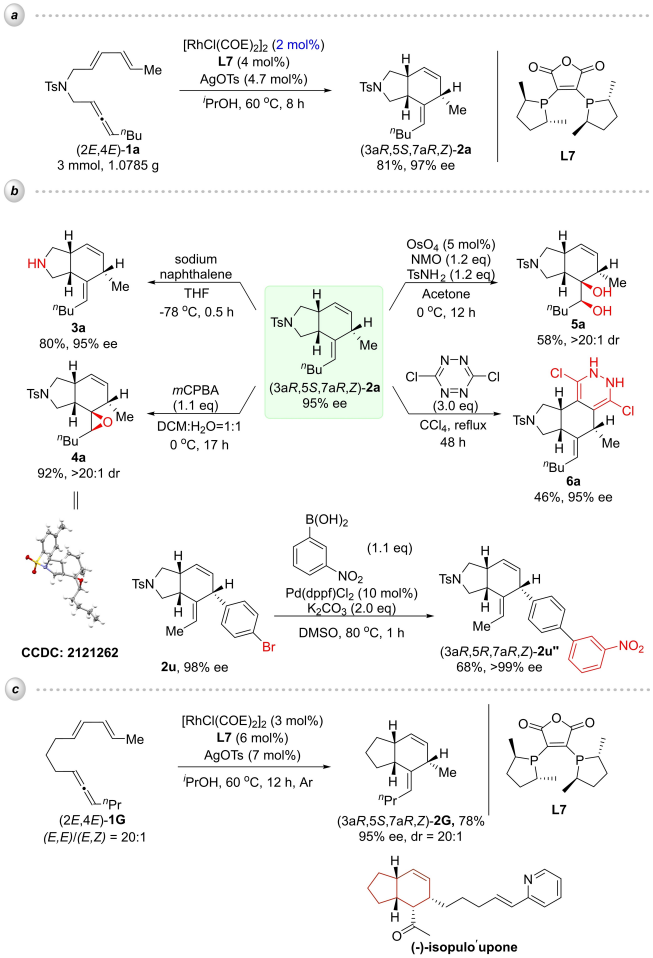
To gain insight into the mechanism of dynamic kinetic process, we prepared both enantiomers of **1a** via EATA (enantioselective allenation of terminal alkynes)[15] and monitored the parallel reactions of *rac*-**1a**, ($S_a$)-**1a** and ($R_a$)-**1a** under standard conditions. All three substrates could undergo [4+2] cycloaddition reaction to afford the same



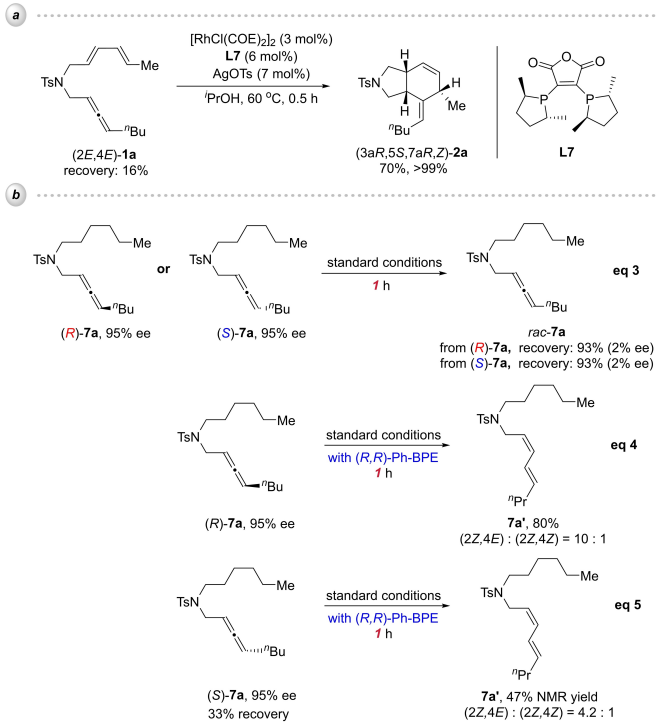

***Scheme 4.*** Rhodium-catalyzed dynamic kinetic intramolecular [4+2] cycloaddition of (2*E*,4*Z*)-**1b** and (2*Z*,4*E*)-**1b**.


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***Scheme 5.*** Gram-scale reaction and synthetic applications.




***Scheme 6.*** Mechanistic studies.

optically active (3a*R*,5*S*,7a*R*,*Z*)-**2a**; ($R_a$)-**1a** had the fastest transformation rate while the reaction of ($S_a$)-**1a** was slowest (Figure 1a and 1b).

During the reactions of ($S_a$)-**1a** or ($R_a$)-**1a**, the ee of the recovered ($S_a$)-**1a** and ($R_a$)-**1a** decreased gradually. When *rac*-**1a** was subjected to the standard conditions, the ee of recovered **1a** increased first then undergo rapid racemization due to the faster reaction rate of ($R_a$)-**1a** (Figure 1c and 1d).

Moreover, (3a*R*,5*S*,7a*R*,*Z*)-**2a** could be obtained with a higher 99 % ee (95 % ee after 0.5 h at 80 °C) with the lower temperature of 60 °C (Scheme 6a). When (*R*)-**7a** or (*S*)-**7a**, in which the C6-1,3-diene unit in **1a** was replaced with *n*-hexyl group, was submitted to the standard conditions, a complete racemization was observed within 1 hour, confirming that the racemization of the allene unit is efficient and enabled by the Rh catalyst (Scheme 6b, eq 3). When (*R*,*R*)-Ph-BPE[4e] was submitted to the standard conditions replacing (*R*,*R*)-Malphos, (*R*)-**7a** was converted to diene **7a**′ with the ratio of 10:1 for (2*Z*,4*E*)-**7a**′ and (2*Z*,4*Z*)-**7a**′ (Scheme 6b, eq 4) while (*S*)-**7a** was partially converted to **7a**′ with the ratio of 4.2:1 (Scheme 6b, eq 5). These data confirmed the striking effect of ligand (Malphos vs. Ph-BPE) on the reaction of the allene unit. In order to test whether such a racemization process with Malphos was catalyzed by the in situ generated Rh–H species, the reaction of (2*E*,4*E*)-**1a** was conducted in aprotic solvent such as DCE running the reaction (Scheme 7a, entry 4), the [4+2]-cycloaddition product (3a*R*,5*S*,7a*R*,*Z*)-**2a** was generated in a low yield with low ee. No reaction was observed in toluene, THF or dioxane (Scheme 7a, entries 1–3). When ($R_a$,2*E*,4*E*)-**1a** was submitted to the standard conditions using toluene or dioxane as the solvent, such a racemization was also observed (Scheme 7b). These results confirmed that the racemization process would most probably not induced by rhodium hydride species. Based on our early study,[4e] the racemization may be caused by the coordination of ($S_a$,2*E*,4*E*)-**1** with the $[RhL^*]^+$ species, which was followed by subsequent C–H oxidative addition and reductive elimination, causing the racemization with (*R*,*R*)-Malphos as the chiral ligand. However, further study is required to completely clarify the nature of the racemization process.

Further scope study revealed that electron-withdrawing $R^1$ group would not promote the isomerization of the endocyclic double bond (Scheme 3, **2x**). However, the previous studies revealed that no reaction occurred in the absence of the ligand while the C=C isomerization product (3a*S*,7a*S*,*E*)-**2a**′ was formed in the presence of (*R*,*R*)-Ph-BPE or (*S*,*S*)-Ph-BPE.[4e] In addition, further control experiments revealed that there was no reaction occurred in the absence of the silver salt (Scheme 8).

Furthermore, isomerization process did not occur when only (*R*,*R*)-Ph-BPE was applied, indicating that the rhodium catalyst, Ph-BPE ligand, and silver salt are all responsible for the isomerization process. We speculated the (*R*,*R*)-


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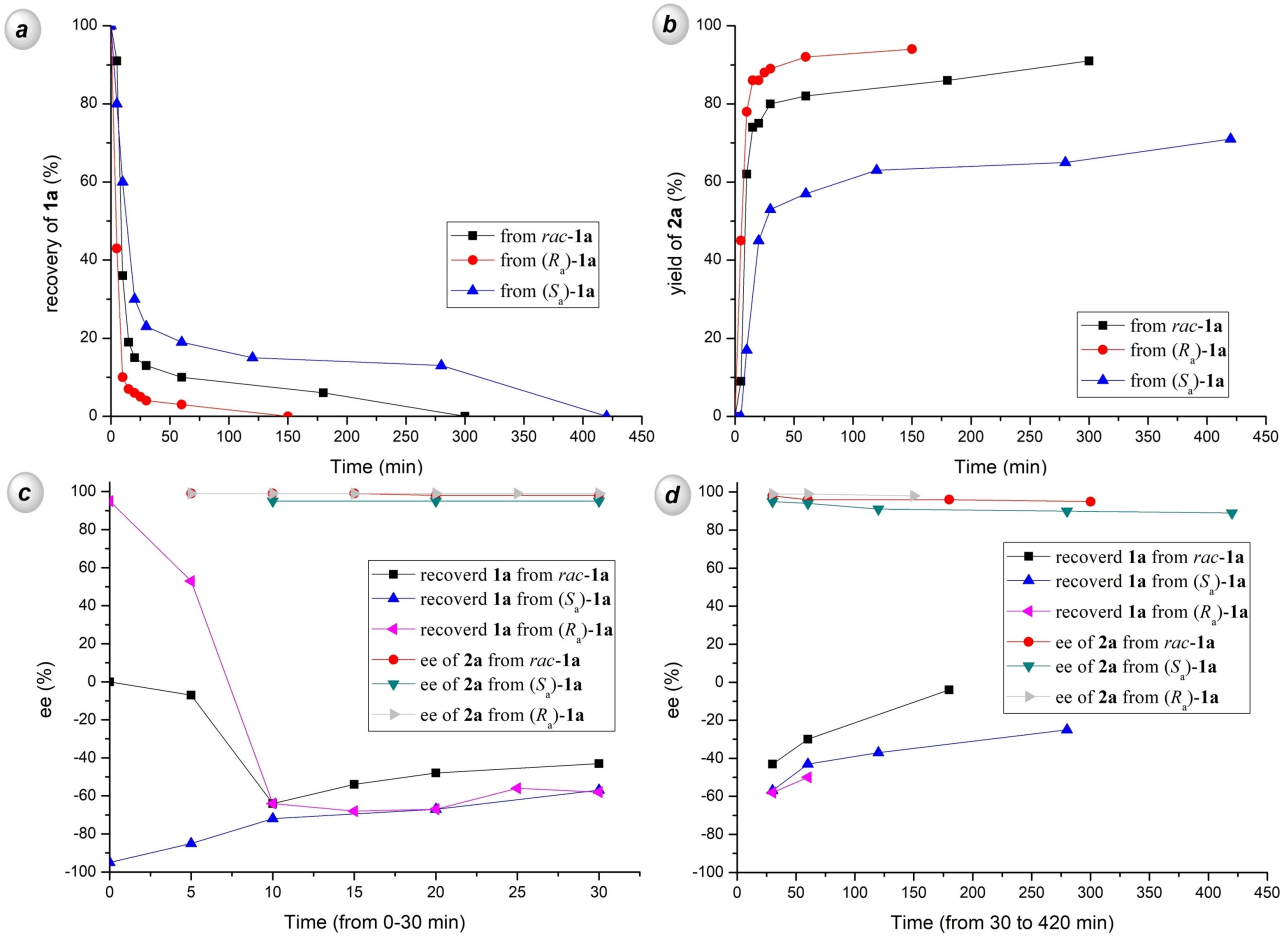



***Figure 1.*** Mechanistic studies. The reaction profile as monitored by $^1$H NMR and HPLC. a) The recovery of **1a**. b) The yield of **2a**. c) The ee of recovered **1a** and product **2a** from 0 min to 30 min. d) The ee of recovered **1a** and product **2a** from 30 min to the end of the reaction.

Malphos is more electron-deficient and its backbone is more rigid, which may make the coordination of the C=C bond in **2a** harder, thus, hinders the C=C bond isomerization process.

In order to rationalize the observed unique stereoselectivity, density functional theory (DFT) calculations have been performed using (2*E*,4*E*)-**1b** as the model substrate (Scheme 9, see the Supporting Information for computational details). The calculation started from the coordination complex **Int1** (selected as the free energy reference), in which the cationic chiral catalyst $[RhL^*]^+$ coordinates to both 1,3-diene and allene unit of ($R_a$,2*E*,4*E*)-**1b**. The cyclometalation of **Int1** is calculated to be exergonic ($\Delta G_{sol} = -18.8$ kcal mol$^{-1}$) and requires an activation barrier of 8.4 kcal mol$^{-1}$ (**TS1**), affording the rhodiabicyclic complex **Int2** reversibly. Subsequent reductive elimination produces the Rh-coordinated product (3a*R*,5*S*,7a*R*,*Z*)-**2b** with a free energy barrier of 22.1 kcal mol$^{-1}$ (**TS2**). On the other hand, the coordination of ($S_a$,2*E*,4*E*)-**1b** with the cationic chiral catalyst $[RhL^*]^+$ generates complex **Int1′**, which is less stable than **Int1** by 6.9 kcal mol$^{-1}$ in free energy. The similar cyclometalation/reductive elimination process is also obtained from **Int1′**, leading to the formation of Rh-coordinated (3a*S*,5*R*,7a*S*,*Z*)-**2b**. However, the cyclometalation from **Int1′** needs to overcome a free energy barrier of 17.2 kcal mol$^{-1}$ (**TS1′**), which is less favorable than **TS1** by 8.8 kcal mol$^{-1}$. The calculated energy difference of 8.8 kcal mol$^{-1}$ accounts for the experimental ee value of over 99 % for substrate (2*E*,4*E*)-**1b**.

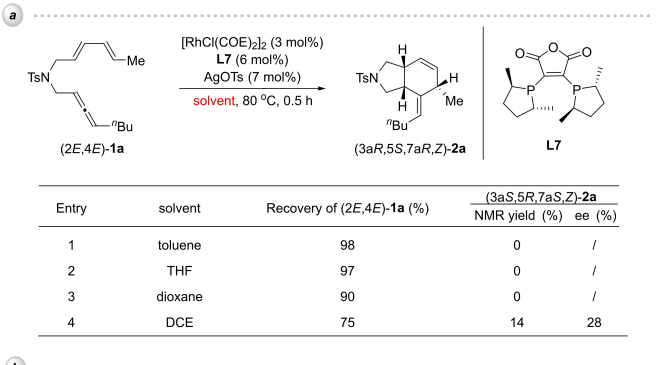
We further tried the intermolecular [4+2]-cycloaddition reaction with (*E*,*E*)-hexadienyl amine **8a** and 1,3-disubstituted allene **9a** or monosubstituted allene **9b** under standard conditions. Unfortunately, no cyclic products were afforded (Scheme 10).

## Conclusion

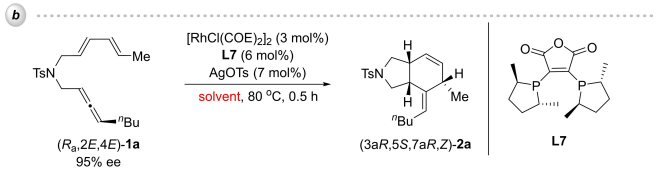
In conclusion, we have developed the Rh-catalyzed dynamic kinetic intramolecular [4+2]-cycloaddition reaction of 1,3-disubstituted allene-dienes to afford *cis*-fused [4.3.0]bicyclic products generating three chiral centers in one shot with an excellent diastereoselectivity and enantioselectivity by applying Malphos as the chiral ligand. The reaction exhibits good substrate scope and functional group compatibility. The synthetic potential of the products for the asymmetric syntheses of other useful building blocks has also been demonstrated. Mechanistic studies demonstrated the nature


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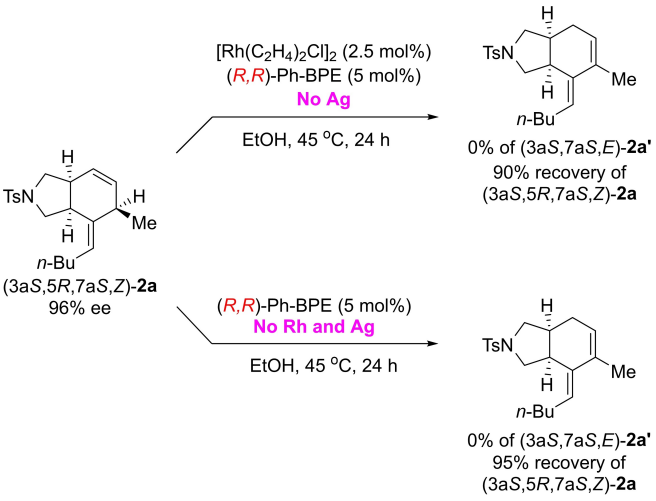



| Entry | solvent | Recovery of (2*E*,4*E*)-**1a** (%) | (3a*S*,5*R*,7a*S*,*Z*)-**2a** NMR yield (%) | ee (%) |
|---|---|---|---|---|
| 1 | toluene | 98 | 0 | / |
| 2 | THF | 97 | 0 | / |
| 3 | dioxane | 90 | 0 | / |
| 4 | DCE | 75 | 14 | 28 |




| Entry | solvent | NMR yield of (3a*S*,5*R*,7a*S*,*Z*)-**2a** (%) | (*Ra*,2*E*,4*E*)-**1a** recovery (%) | ee (%) |
|---|---|---|---|---|
| 1 | toluene | 0 | 90 | 40 |
| 2 | dioxane | 0 | 97 | 24 |

***Scheme 7.*** Racemization studies. Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a** or (*Ra*,2*E*,4*E*)-**1a**, 3 mol% of $[RhCl(COE)_2]_2$, 6 mol% of **L7**, and 7 mol% of AgOTs in solvent (1 mL) at 80 °C. The yield and recovery were determined by $^1$H NMR analysis using mesitylene as the internal standard. The ee was determined by HPLC analysis.

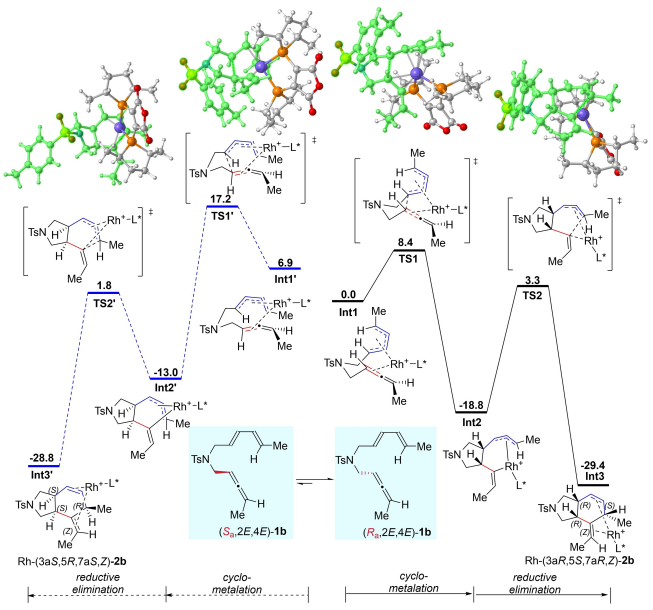



***Scheme 8.*** Control experiments.

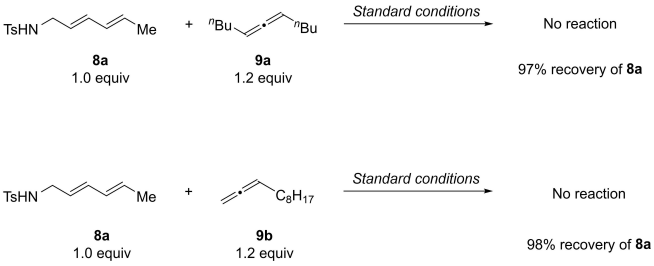
of the dynamic kinetic process. Further studies related to this topic including the exact mechanism of the allene racemization are in progress in our laboratory.

## Acknowledgements

Financial support from National Natural Science Foundation of China (Grant No. 21988101 for S. Ma and Grant No. 21801041 for H. Qian) are greatly appreciated. We are very grateful to Professor Armin Börner for providing us with Malphos. We also thank Mr. Yulong Song in this group for reproducing the results of (3a*R*,5*S*,7a*R*,*Z*)-**2f** and (3a*R*,5*R*,7a*R*,*Z*)-**2s** presented in this study.




***Scheme 9.*** Free energy profiles calculated for the cyclometalation/reductive elimination process from **Int1** and **Int1'** at B3LYP-D3(BJ)/6-311+G(d,p)-SDD,SMD(iPrOH)//B3LYP-D3(BJ)/6-31G(d,p)-LANL2DZ level. Relative free energies are given in kcal mol$^{-1}$ with respect of **Int1**.




***Scheme 10.*** Intermolecular [4+2]-cycloaddition reaction.

## Conflict of Interest

The authors declare no conflict of interest.

## Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

**Keywords:** Allenes · Asymmetric Catalysis · Cycloaddition · Rhodium

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Manuscript received: August 7, 2022
Accepted manuscript online: September 21, 2022
Version of record online: October 21, 2022



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